

RNase cleavage products in mzLib

Succinct biological reference and implementation-validation report | scope: all RNases identified during the review

Purpose. Establish the cleavage context, terminal chemistry, and evidence strength needed to validate the RNase definitions and digestion products. Frequencies below are reported site efficiencies or occurrence behavior when the source provides them; otherwise the report says explicitly that no universal rate is established.

Scope boundary. Part I includes only RNases with reported ssRNA cleavage activity, including condition-specific PhyM rows and separate mammalian RNase 1 and E. coli RNase I entries. The Part II text block retains the 14 rnases2.tsv configuration records, including non-biological search modes.

Nomenclature. RNase 1 and RNase I are distinct: mammalian RNase 1 is a pancreatic-type pyrimidine-preferring enzyme; E. coli RNase I is a largely sequence-independent nuclease.

Formula key

Field	Remainder	Meaning at a newly created terminus	Evidence
3' product	H ₂ O ₄ P	Linear 3'-phosphate default	E001,E003
3' product	O ₃ P	2',3'-cyclic-phosphate remainder; H ₂ O ₄ P + H-2 O-1	E003,E006,E007
5' product	O-3P-1	5'-OH remainder (removes phosphate)	E001,E003,E006
Blank	-	No enzymatic product or terminus not established; safe fallback	E002

Interpretation: the 3' product is the fragment ending at the cut; the 5' product is the fragment beginning immediately after the cut. A blank row value must not be read as a measured zero.

Part I. Biological cleavage and products

Definition	Biological cleavage	mzLib cleavage	3' biological product	3' formula	5' biological product	5' formula	Occurrence / frequency evidence	Sources
RNase T1	Cleaves after G in ssRNA (GpN).	G (full)	3'-P; cyclic intermediate	H2O4P	5'-OH	O-3P-1	All GpN bonds observed in a defined test RNA under optimal digestion; no missed cuts reported	E006,E015
RNase T2	ssRNA endonuclease; little base specificity.	G ,C ,A ,U (full)	3'-P; cyclic intermediate	H2O4P	5'-OH	O-3P-1	Single-strand specific; no strong base preference; no universal rate reported	E007,E025
RNase A	Cleaves after unpaired C or U.	C ,U (full)	3'-P; cyclic intermediate	H2O4P	5'-OH	O-3P-1	C/U sites; C is reported ~2x faster than U in a vendor summary; structure/conditions also matter	E008,E028
RNase 1	Mammalian RNase 1; cleaves ssRNA 3' to pyrimidines and can also act on dsRNA.	G ,C ,A ,U (full)	3'-P; cyclic intermediate	H2O4P	5'-OH	O-3P-1	No universal site rate reported; RNase 1 is present in vertebrate pancreas and is secreted by human vascular endothelium	E030,E031
RNase I	E. coli RNase I; largely sequence-independent ssRNA endonuclease; Ca2+ can support dsRNA activity.	G ,C ,A ,U (not modeled)	3'-P; cyclic intermediate	O3P	5'-OH	O-3P-1	Largely sequence-independent cleavage; no universal site rate reported	E009
RNase PhyM (7 M urea)	Cleaves Up-N and Ap-N under 7 M urea, 50 °C, pH 5.	A ,U (full)	2',3'-cyclic-P observed; 5'-OH not reported	O3P	Unknown in cited source	—	Almost exclusive and uniform at Up-N/Ap-N under the stated sequencing conditions; no percentage reported	E010,E029
RNase PhyM (no urea)	Cleaves Up-N, Ap-N and Gp-N without urea.	A ,U ,G (full)	2',3'-cyclic-P observed; 5'-OH not reported	O3P	Unknown in cited source	—	Gp-N is additionally attacked without urea; relative efficiencies not reported	E010,E029
RNase 4	Cleaves after U in U-A and U-G ssRNA sites.	U A,U G (full)	Mixed: linear 3'-P and 2',3'-cyclic-P	H2O4P,O3P	5'-OH	O-3P-1	UA is slightly preferred over UG; no absolute site percentage reported	E011,E018
RNase U2	Purine-selective cleavage after A or G; A preferred.	G ,A (full)	3'-P; cyclic intermediate	H2O4P	5'-OH	O-3P-1	A > G (reported order A > G >> C > U); no universal rate	E012,E013

Definition	Biological cleavage	mzLib cleavage	3' biological product	3' formula	5' biological product	5' formula	Occurrence / frequency evidence	Sources
Cusativin	Prefers CpA, CpG and CpU; CpC is resistant; some UpA/UpU/ApU cleavage occurs.	C (full)	Mixed: linear 3'-P and 2',3'-cyclic-P	H2O4P,O3P	5'-OH	O-3P-1	CpA 70-90%; CpG/CpU 80-100%; UpA 70-90%; UpU/ApU 14-30%; CpC 0% in the LC-MS study	E014,E015
RNase_MC1	Cleaves at the 5' side of U; ApU/CpU/UpU preferred and GpU nearly resistant.	U (full)	Mixed: linear 3'-P and 2',3'-cyclic-P	H2O4P,O3P	5'-OH	O-3P-1	ApU 70-100%; CpU 80-100%; UpU 45-60%; GpU 1.0-1.5%; CpA 10-20%	E015
colicin_E5	Sequence/structure-specific tRNase at anticodon positions 34-35; Q-U/G-U context.	G U (full)	2',3'-cyclic-P	O3P	5'-OH	O-3P-1	Cleaves susceptible tRNAs at positions 34-35; not transcript-wide	E016
RNase CL3	Chicken-liver cytidine-preferential endoribonuclease.	C (not modeled)	Not established in cited source	—	Not established in cited source	—	C-containing to U-containing bond discrimination was 61:1 in limit digests of 5S/5.8S rRNA	E024
RNase L	2-5A-activated, metal-independent ssRNA endoribonuclease; predominantly after U in UA/UU.	not modeled	2',3'-cyclic-P	O3P	5'-OH	O-3P-1	Preference approximately UU > UA >> UG > UC; cleavage magnitude depends strongly on local structure	E019,E026
RNase E	Low-specificity cleavage in AU-rich ssRNA; 5'-monophosphate-assisted pathway.	not modeled	Not established in cited source	—	Not established in cited source	—	Prefers >=3 unpaired 5' nucleotides; optimal spacing ~8 nt; U/A/G-rich constructs diffused 2-3x faster than C-rich RNA in one assay	E023,E027
RNase G	RNase E homolog; low-specificity cleavage in AU-rich ssRNA; 5'-monophosphate-assisted pathway.	not modeled	Not established in cited source	—	Not established in cited source	—	Prefers >=3 unpaired 5' nucleotides; optimal spacing ~6 nt; generally more impeded than RNase E with <4 unpaired nt	E023

Frequency note: a percentage is reported only where a cited study measured one. Otherwise, the table gives the observed preference, spacing/structure requirement, or an explicit evidence gap; none is a universal rate constant. MC1 and Cusativin percentages are study-specific efficiencies.

Part II. mzLib implementation and resulting products

The following is the machine-readable text representation used for direct comparison with Part I. It is copied from rnases2.tsv; blank remainder cells are intentional.

```
Name\tMotif\tSpecificity\t3' remainder\t5' remainder\tNotes
RNase T1\tG\tfull\tH2O4P\tO-3P-1\tG, ssRNA; src=S1
RNase T2\tG,C,A,U\tfull\tH2O4P\tO-3P-1\tssRNA, little base specificity; all-base motif is approximate; src=S2
RNase A\tC\tU\tfull\tH2O4P\tO-3P-1\tC/U, ssRNA; src=S3
top-down\tnone\t\t\t\tNo enzymatic cleavage; retains input termini; src=L1
RNase 1\tG,C,A,U\tfull\tH2O4P\tO-3P-1\tMammalian RNase 1; pyrimidine-preferring biology; existing all-base motif is broader; src=S13,S14
RNase PhyM (7 M urea)\tA,U\tfull\tO3P\tUp-N/Ap-N; almost exclusive and uniform under 7 M urea; 3' cyclic-P observed; 5' terminus not established; src=S5,S12
RNase PhyM (no urea)\tA,U,G\tfull\tO3P\tUp-N/Ap-N/Gp-N; no-urea condition; 3' cyclic-P observed; 5' terminus not established; src=S5,S12
RNase 4\tU,A,U,G\tfull\tH2O4P,O3P\tO-3P-1\tU-A slightly preferred over U-G; ssRNA; mixed 3' products; src=S6,S11
RNase U2\tG,A\tfull\tH2O4P\tO-3P-1\tPurines, A preferred over G; motif does not encode weighting; src=S7
Cusativin\tC\tfull\tH2O4P,O3P\tO-3P-1\tCpA 70-90%; CpG/CpU 80-100%; CpC resistant; mixed 3' products; motif simplified; src=S8,S9
RNase_MC1\tU\tfull\tH2O4P,O3P\tO-3P-1\tApU 70-100%; CpU 80-100%; UpU 45-60%; GpU 1.0-1.5%; mixed 3' products; motif simplified; src=S9
colicin_E5\tG\tfull\tO3P\tO-3P-1\tQ-U/G-U anticodon-loop cleavage; current G/U motif approximate; src=S10
singleN\t\tSingleN\tH2O4P\tO-3P-1\t5'-anchored search ladder; internal boundaries use these defaults; src=L1
singleC\t\tSingleC\tH2O4P\tO-3P-1\t3'-anchored search ladder; internal boundaries use these defaults; src=L1
```

The literal \t markers show the six tab-delimited fields: Name, Motif, Specificity, 3' remainder, 5' remainder, and Notes. Comma-separated remainder formulas are alternative products, not probabilities; the reader should split them into list-valued termini. Vector text/tables remain resolution-independent in the PDF.

Implementation checks

- Data contract.** rnases2.tsv preserves Name, Motif, and Specificity, and appends ThreePrimeTerminusRemainder, FivePrimeTerminusRemainder, and Notes. Formula values are remainder formulas, not descriptive labels. E002
- Rnase API.** Rnase.cs exposes IList-valued ThreePrimeTerminusRemainder and FivePrimeTerminusRemainder. Optional constructor arguments default to singleton lists containing H2O4P and O-3P-1, preserving previous behavior when omitted. E001
- Boundary assignment.** For an internal product, the fragment ending at the cleavage site receives the configured 3' remainder and the fragment beginning after it receives the configured 5' remainder. Original RNA termini are retained when the product includes the corresponding source end. E001
- Cyclic product.** O3P is the repository-compatible cyclic remainder derived from the existing H-2 O-1 cyclic-phosphate modification. E003
- Runtime integration status.** The current RnaseDictionary parser resolves only Name, Motif, and Specificity and constructs Rnase without the two formula arguments. Therefore rnases2.tsv documents the intended metadata but is not yet the active source of per-row terminus formulas. E004
- Multiple products.** For an internal product, Rnase.cs enumerates the Cartesian product of all configured 3' and 5' remainder lists. RNase 4 is encoded as H2O4P,O3P; the reader should split that cell into two alternatives. This records alternatives, not probabilities. E001,E002
- Expected standard digest.** With the current defaults, each internal cleavage produces a 3'-phosphate remainder H2O4P on the upstream fragment and a 5'-OH remainder O-3P-1 on the downstream fragment. E001
- Special rows.** top-down retains input termini; singleN and singleC generate deterministic anchored search ladders and use the configured defaults for internal boundaries. E001,E002
- TSV text representation.** The Part II text-format block is copied from rnases2.tsv so comma-separated formula alternatives can be compared directly with the combined table. E002
- PhyM rows.** PhyM is represented as two condition-specific rows: A/U at 7 M urea and A/U/G without urea. The sources support a 3' cyclic-phosphate product assignment, but do not report the 5' terminus or numeric efficiencies. E002,E010,E029

Worked product mapping

For an internal cut in A-C|G-U, the expected default pair is: upstream A-C with 3'-remainder H_2O_4P ; downstream G-U with 5'-remainder O-3P-1. If the RNase definition supplies O_3P , only the upstream 3' remainder changes. The source RNA's original 5' and 3' termini are not replaced when retained by the product. E001,E003

Multiple-terminus example: RNase 4 can be represented as $ThreePrimeTerminusRemainder = [H_2O_4P, O_3P]$ and $FivePrimeTerminusRemainder = [O-3P-1]$. An internal cut then yields two upstream-product formula alternatives; multiple values on both sides yield their Cartesian combinations. This records alternatives, not probabilities. E001,E011,E018

Validation status

- Definition fidelity: rnases2.tsv retains the original definitions and splits PhyM into two condition-specific rows; it contains 14 records. Five additional ssRNA-capable literature-only RNases are included in Part I for validation scope. E002
- Formula fidelity: Rnase.cs defaults match the local formula key and RnaMods.txt entries. E001,E003,E005

- Multiplicity fidelity: the changed IList API and nested enumeration support multiple 3' and/or 5' remainder formulas, including Cartesian combinations. E001
- Product behavior covered by existing tests: terminal formulas, cyclic 3' digestion products, and terminal-modification mass behavior. E005
- Integration gate: before claiming per-RNase products are active, update or verify RNaseDictionary parsing of the two optional formula columns and add a test that loads rnases2.tsv. E004

Limitations

Motif syntax does not encode secondary structure, quantitative cleavage weighting, enzyme activation, or mixed-product distributions. RNase.cs can represent alternatives, and rnases2.tsv now uses comma-separated formula cells for RNase 4, Cusativin, and MC1; reader integration is pending. PhyM specificity/frequency is condition-specific, and its 3' cyclic product is supported by MALDI-MS, but its 5' chemistry and numeric efficiencies remain unreported. Literature-only RNases require separate modeling decisions. E002,E004,E010,E011,E014,E015,E029

References and provenance

Source IDs are used in the tables so each claim can be traced to the record below. Web sources were retrieved for drafting; human source verification, including exact locator checking, remains required before scientific sign-off.

- E001.** mzLib RNase.cs (local implementation). 2026. E:/agentplayground/mzLib/mzLib/Transcriptomics/Digestion/RNase.cs. Locator: lines 9-21, 73-240.
- E002.** mzLib rnases2.tsv (local definition table). 2026. E:/agentplayground/mzLib/mzLib/Transcriptomics/Digestion/rnases2.tsv. Locator: lines 1-54.
- E003.** mzLib RnaMods.txt (local terminal modifications). 2026. E:/agentplayground/mzLib/mzLib/Omics/Resources/RnaMods.txt. Locator: lines 155-172.
- E004.** mzLib RNaseDictionary.cs (local TSV parser). 2026. E:/agentplayground/mzLib/mzLib/Transcriptomics/Digestion/RNaseDictionary.cs. Locator: lines 260-305.
- E005.** mzLib transcriptomics tests (local validation). 2026. E:/agentplayground/mzLib/mzLib/Test/Transcriptomics/. Locator: TestOsmReading.cs 224-240; TestDigestion.cs 351-417; TestFragmentation.cs 359-410.
- E006.** ENZYME EC 4.6.1.24: ribonuclease T1. n.d.. <https://enzyme.expasy.org/EC/4.6.1.24>. Locator: Accepted Name, Reaction catalysed, Comment(s).
- E007.** ENZYME EC 4.6.1.19: ribonuclease T2. n.d.. <https://enzyme.expasy.org/EC/4.6.1.19>. Locator: Accepted Name, Reaction catalysed, Comment(s).
- E008.** Thermo Fisher Scientific: Deciphering Nuclease Digestion Data. n.d.. <https://www.thermofisher.com/us/en/home/references/ambion-tech-support/rna-structure-function/tech-notes/deciphering-nuclease-digestion-data.html>. Locator: RNase A and RNase T1 specificity discussion.
- E009.** E. coli RNase I exhibits a strong Ca²⁺-dependent inherent double-stranded RNase activity. 2021. <https://doi.org/10.1093/nar/gkab284>. Locator: Nucleic Acids Research 49(9), 5265-5278; abstract/introduction.
- E010.** Donis-Keller (1980): Phy M: an RNase activity specific for U and A residues useful in RNA sequence analysis. 1980. <https://pmc.ncbi.nlm.nih.gov/articles/PMC324360/>. Locator: Abstract.
- E011.** NEB M1284: RNase 4. n.d.. <https://www.neb.com/en-us/products/m1284-rnase-4>. Locator: Product description.
- E012.** Improving RNA Modification Mapping Sequence Coverage by LC-MS through a Nonspecific RNase U2-E49A Mutant. 2018. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6214470/>. Locator: Introduction: RNase U2 specificity.
- E013.** ENZYME EC 4.6.1.20: ribonuclease U2. n.d.. <https://enzyme.expasy.org/EC/4.6.1.20>. Locator: Reaction catalysed, Comment(s).
- E014.** Novel ribonuclease activity of cusativin from Cucumis sativus for mapping nucleoside modifications in RNA. 2017. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5709141/>. Locator: Abstract and Introduction.
- E015.** RNA Cleavage Properties of Nucleobase-Specific RNase MC1 and Cusativin. 2022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9266746/>. Locator: Abstract and Results 2.1.1.
- E016.** Structural basis for sequence-dependent recognition of colicin E5 tRNase. 2006. <https://pmc.ncbi.nlm.nih.gov/articles/PMC1669751/>. Locator: Abstract and cleavage-product discussion.
- E017.** Kassiss et al. (2026): Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents. 2026. <https://doi.org/10.48550/arXiv.2609.00065>. Locator: arXiv record, latest version.
- E018.** Human RNase 4 improves mRNA sequence characterization by LC-MS/MS. 2022. <https://academic.oup.com/nar/article/50/18/e106/6649373>. Locator: Figure 1D and discussion of UA versus UG cleavage.
- E019.** Ribonuclease L and metal-ion-independent endoribonuclease cleavage sites in host and viral RNAs. 2014. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4005677/>. Locator: Abstract, Introduction, and Table 1.
- E023.** Distinct Requirements for 5'-Monophosphate-assisted RNA Cleavage by Escherichia coli RNase E and RNase G. 2016. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4777840/>. Locator: Abstract and Introduction.
- E024.** The purification and properties of chicken liver RNase: An enzyme which is useful in distinguishing between cytidylic and uridylic acid residues. 1980. <https://pubmed.ncbi.nlm.nih.gov/6243659/>. Locator: Abstract: 61:1 cytidylic-to-uridylic bond discrimination.
- E025.** Several RNase T2 enzymes function in induced tRNA and rRNA turnover in the ciliate Tetrahymena. 2012. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3248902/>. Locator: Abstract: single-strand specificity and lack of strong base preference.
- E026.** New advances in our understanding of the unique RNase L in host pathogen interaction and immune signaling. 2020. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7128181/>. Locator: RNase L cleavage-preference discussion.
- E027.** RNase E searches for cleavage sites in RNA by linear diffusion: direct evidence from single-molecule FRET. 2024. <https://academic.oup.com/nar/article/52/11/6674/7655786>. Locator: Abstract/results: cleavage-rate comparison for U/A/G versus C-rich RNA.

E028. RNase A: Frequently Asked Questions. n.d.. <https://agscientific.com/blog/rnase-a-faq.html>. Locator: Cleavage chemistry and reported two-fold C-versus-U rate difference.

E029. Sequencing RNA by a combination of exonuclease digestion and uridine specific chemical cleavage using MALDI-TOF. 1998. <https://academic.oup.com/nar/article/26/2/446/1170657>. Locator: RNase PhyM digest; MALDI-TOF products assigned 2',3'-cyclic phosphate.

E030. Nucleotide substrate binding characterization in human pancreatic-type ribonucleases. 2019. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0220037>. Locator: Abstract and Introduction: hRNase 1-7 substrate preferences and catalytic-efficiency range.

E031. Pancreatic Ribonuclease. 2018. <https://pancreapedia.org/molecules/pancreatic-ribonuclease>. Locator: RNASE1 entry: ssRNA pyrimidine cleavage and vertebrate-pancreas occurrence.

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